

cw-college-admissions

March 3, 2026

1 Classwork : The Admissions Black Box

Below is a 30-minute activity called “**The Admissions Black Box.**” It uses the lesson’s Circle/Square analogy to show how bias enters a system and how hard it is to “fix” it without losing accuracy.

1.1 1. Setup (5 Minutes)

Divide the class into small groups (3–4 students). Each group is an **Admissions Committee** for “Polygon University.”

- **The Data:** Give each group a stack of 10 “Applicant Cards.” (or share list)
- **Circles:** Mostly have high SAT scores (1400+) and 3+ extracurriculars.
- **Squares:** Mostly have mid-range SAT scores (1100–1200) and 1–2 extracurriculars.
- **The Secret “Ground Truth”:** Tell the students: On this planet, Squares have no access to SAT tutors or expensive sports, but they are equally “capable” of succeeding in college.

1.2 2. Phase 1: Human Bias (10 Minutes)

Ask the groups to quickly select 4 students to admit based **only** on the SAT score and extracurriculars.

- **The Result:** Most groups will admit 4 Circles.
- **Discussion:** Ask: “Is this fair? You picked the ‘best’ scores, but did you pick the ‘best’ potential?”

1.3 3. Phase 2: The “Blind” Algorithm (5 Minutes)

Tell them they are now training an AI. They must create a simple “Rule” (an algorithm) for admission.

- **Constraint:** They are **not allowed** to use the “Shape” (Race) variable.
- **Common Rule:** “*Admit if SAT > 1300.*”
- **The Reveal:** Show them that because of the zip codes or school names on the cards (which correlate with the shapes), their “blind” rule still only admits Circles.
- **Lesson:** Removing the label doesn’t remove the bias.

1.4 4. Phase 3: The Fairness Trade-off (10 Minutes)

Now, tell the groups they must comply with the “**Fairness Act of 2026.**” They must ensure at least 1 Square is admitted.

- **The Conflict:** To admit a Square, they have to reject a Circle with a higher SAT score.
- **The “Pareto” Moment:** Ask the groups:
- “Does your university’s ‘average SAT’ go down? (Accuracy Loss)”
- “Is it ‘fair’ to the Circle you rejected? (Individual Fairness)”
- “Is it ‘fair’ to the Square community to ignore their lack of tutors? (Group Fairness)”

1.5 Wrap-up Discussion Questions

- **The “Why”:** If we know the SAT is a biased measure for Squares, should we keep using it?
- **Generative AI:** If an AI wrote the “Acceptance Letters,” and it used formal, flowery language for Circles but “simplified” language for Squares because of their data profile, would that be a problem? (This connects to the **Representational Harm** update in your lesson).
- **The Future:** How would a “Human-in-the-loop” (a real person checking the AI’s math) have changed the outcome?

2 Materials

2.1 1. Student Handout: Polygon University Applicant Pool

Copy this into your student-facing materials or show it on the screen.

ID	Race	SAT Score	Extracurricular Activities
C1	Circle	1510	Varsity Tennis, Cello, 50 Volunteer Hours
C2	Circle	1480	Debate Team Captain, Intern at Tech Co.
C3	Circle	1420	Sailing Club, AP Scholar
C4	Circle	1390	Chess Club, Private Piano Lessons
C5	Circle	1350	Lacrosse, Summer Abroad in France
S1	Square	1250	Part-time Job (20hrs/wk), Raised 3 younger siblings
S2	Square	1210	School Band, Community Center Tutor
S3	Square	1180	Self-taught Coding, Art Club
S4	Square	1150	Varsity Football, Worked at family shop
S5	Square	1090	Part-time Job (30hrs/wk)

2.2 2. Teacher’s Master Key: The “Ground Truth”

The final reveal to show the “Construct Space” vs. the “Observed Space.”

ID	Race	SAT	True Potential (Secret)	The “Why” (Context)
S1	Square	1250	Extremely High	High grit; score is low only due to lack of study time.
C1	Circle	1510	High	Highly capable; benefited from \$5k SAT prep course.

ID	Race	SAT	True Potential (Secret)	The “Why” (Context)
S2	Square	1210	High	Strong leadership; limited resources at school.
S3	Square	1180	High	Brilliant coder; English as a second language.
C2	Circle	1480	Medium	Good student; scores reflect heavy parental support.
C3	Circle	1420	Medium	Average effort; high-quality school district.
S4	Square	1150	Medium	Average capability; worked hard despite obstacles.
C4	Circle	1390	Low	“Over-tutored”; reached their ceiling in high school.
C5	Circle	1350	Low	Reliant on external help to maintain grades.
S5	Square	1090	Medium	High work ethic; SAT is unreliable for this student.

2.3 Classroom Debrief: Tying it to 2026 Regulations

1. **The “High Risk” Label:** Under the **EU AI Act**, this admissions model is “High Risk.” If your school used it, you would have to prove you checked for the bias that just occurred in the classroom.
2. **Individual vs. Group Fairness:** If we admit **S1** over **C4**, we are being “Group Fair” (correcting for systemic bias) but **C4** might complain that it’s “Individually Unfair” because their SAT score was higher. There is no easy math to solve this - it’s a policy decision.
3. **The Human in the Loop:** In 2026, we don’t let the AI make the final “Deny.” We use the AI to flag potential, but a human must look at the “Context” column (the “Why”) to make the final call.